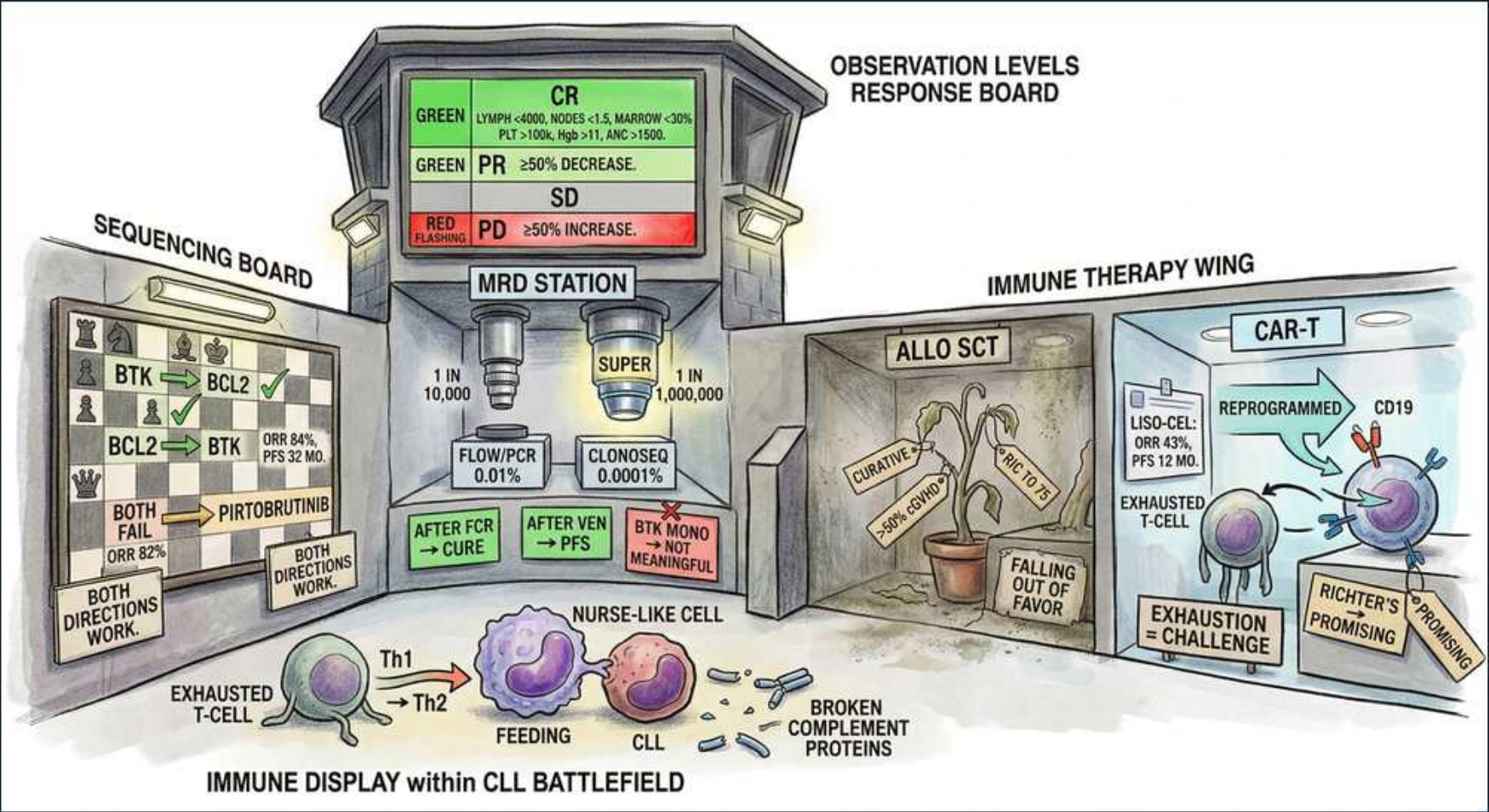


CLL Response Assessment, MRD & Modern Therapy



1. Response board (CR/PR/SD/PD)

WHAT YOU SEE

Green CR/PR, yellow SD, red flashing PD on an 'observation levels' board

WHAT IT ENCODES

iwCLL response criteria. CR = lymphocytes normal, nodes <1.5 cm, marrow <30% lymphocytes, platelets >100k, Hgb >11, neutrophils >1.5k. PR = >=50% drop in lymphocytes/nodes. PD = >=50% increase in nodes/lymphocytes or new disease.

BOARD TRAP

Treatment-induced lymphocytosis from BTK inhibitors (redistribution) is NOT progression — don't call PD when isolated lymphocytosis rises on a BTKi.

2. BTK / BCL2 sequencing board

WHAT YOU SEE

A chessboard showing BTK then BCL2, BCL2 then BTK — both directions work

WHAT IT ENCODES

Sequencing of targeted agents: BTK inhibitor then venetoclax (BCL2), or venetoclax then BTKi — both directions give good ORR (84%, 82%) and durable PFS. Order can be driven by toxicity/comorbidity rather than efficacy.

BOARD TRAP

Don't assume only one sequence works; cross-resistance is incomplete, so a BTKi failure does NOT preclude later venetoclax response and vice versa.

3. Pirtobrutinib (both-fail rescue)

WHAT YOU SEE

Pirtobrutinib labeled where 'both fail'

WHAT IT ENCODES

Pirtobrutinib is a non-covalent (reversible) BTK inhibitor active after both covalent BTKi and venetoclax fail, including in C481S-mutated disease. ORR ~82% in pretreated patients.

BOARD TRAP

C481S mutation causes resistance to COVALENT BTKi (ibrutinib/acala/zanu) but NOT to non-covalent pirtobrutinib — that's the discriminator.

4. MRD station (10⁻⁴ / 10⁻⁶)

WHAT YOU SEE

Microscope station with '1 in 10,000' and a SUPER '1 in 1,000,000' sensitivity

WHAT IT ENCODES

Minimal residual disease (uMRD) measured by flow (10⁻⁴, 0.01%) or clonoSEQ NGS (10⁻⁶, 0.0001%). uMRD predicts longer PFS and enables fixed-duration venetoclax-based therapy.

BOARD TRAP

MRD after BTK monotherapy is NOT meaningful — BTKi is given continuously and rarely achieves uMRD; MRD-guided stopping applies to venetoclax-based fixed-duration regimens.

5. After venetoclax = PFS benefit

WHAT YOU SEE

Green sign 'after VEN = PFS'

WHAT IT ENCODES

Venetoclax-based fixed-duration therapy (e.g. VenO) achieves deep uMRD; depth of MRD correlates with prolonged progression-free survival, supporting time-limited treatment.

BOARD TRAP

Achieving CR alone is less predictive than achieving uMRD; a CR with detectable MRD relapses sooner than a uMRD partial response.

6. Allo SCT (falling out of favor)

WHAT YOU SEE

A wilting potted plant 'falling out of favor', curative but high GVHD

WHAT IT ENCODES

Allogeneic stem cell transplant is the only potentially curative option but carries high transplant-related mortality/GVHD. Reserved for young fit patients with double-refractory (BTKi + venetoclax) high-risk disease.

BOARD TRAP

Allo SCT is NOT first-line; in the targeted-therapy era it is salvage for double-refractory or Richter's, not routine consolidation.

7. CAR-T wing (liso-cel, CD19)

WHAT YOU SEE

Reprogrammed T cells targeting CD19, 'promising'

WHAT IT ENCODES

CD19-directed CAR-T (liso-cel, lisocabtagene maraleucel) is approved for CLL after BTKi and venetoclax failure; ORR ~42%, PFS ~12 months. Promising in heavily pretreated and Richter's settings.

BOARD TRAP

CAR-T targets CD19; it is salvage after both targeted classes fail — not an early-line therapy and response is lower than in large B-cell lymphoma.

8. Exhausted T-cell challenge

WHAT YOU SEE

A tired T-cell representing T-cell exhaustion

WHAT IT ENCODES

CLL induces T-cell exhaustion (PD-1/immune dysfunction), which limits autologous CAR-T efficacy — a key biologic barrier explaining lower CAR-T response rates than in DLBCL.

BOARD TRAP

The low CAR-T efficacy in CLL is due to intrinsic T-cell exhaustion, not tumor antigen loss — don't attribute failure to CD19-negativity alone.

9. Richter's = promising target

WHAT YOU SEE

Sign 'Richter's promising' near CAR-T

WHAT IT ENCODES

Richter's transformation (CLL to DLBCL) is aggressive and chemo-resistant; CAR-T and allo-SCT are being explored as the most promising approaches in fit patients.

BOARD TRAP

Richter's is a clonally-related DLBCL with poor prognosis — standard CLL targeted agents are largely ineffective once transformation occurs.

10. Nurse-like cells feeding CLL

WHAT YOU SEE

Macrophage-like 'nurse-like cell' feeding a CLL cell with Th1/Th2

WHAT IT ENCODES

Nurse-like cells (monocyte-derived) and Th2-skewed T-helper cells in the microenvironment provide survival signals (via BCR, CD40L, BAFF) that sustain CLL — the basis for microenvironment-targeted therapy.

BOARD TRAP

CLL survival is microenvironment-dependent; this is why BCR/BTK signaling blockade works — not because CLL is intrinsically proliferative like acute leukemia.

11. Broken complement proteins

WHAT YOU SEE

Broken/scattered complement protein fragments

WHAT IT ENCODES

Hypocomplementemia and hypogammaglobulinemia in CLL drive immune deficiency; infections are the leading cause of death. Justifies IVIG for recurrent infection with IgG <300-500.

BOARD TRAP

Infection (not bleeding or transformation) is the leading cause of death in CLL; immune paresis is from hypogammaglobulinemia + complement dysfunction.
